# Project Components

## 🧠1️⃣Project Core Architecture Overview

Project Type:

**Predictive + Analytical Web-based Decision Support System**

High Level Flow:

Data → Preprocessing → Risk Calculation → ML Prediction → Storage → Flask API → Dashboard → Admin View

---

## 📐2️⃣MATHS – Exactly Kya Karna Hoga

Maths overkill nahi karna. Targeted maths.

### ✅ A. Risk Score Formula

Define:

$R = w_1T + w_2A + w_3W + w_4P$

Where:

- T = traffic density (normalized)
- A = accident rate
- W = weather severity
- P = pollution index

Must do things :

- Normalization (Min-Max scaling formula)
- Weight justification (domain logic ya data-based correlation)

- Equal weights (baseline)
- Correlation-based weighting
- Regression coefficient-based weighting

---

### ✅ B. Monthly Analysis

- Mean risk per month
- Variance per month
- Trend slope (linear regression slope)
- Seasonal peak detection

Formula:

Slope from simple regression:

$\beta_1 = Cov(X,Y) / Var(X)$

### ✅ C. Correlation Analysis

- Pearson correlation

- Multicollinearity check

---

### ✅ D. Prediction Model

Simple regression:

- Linear Regression
- Ridge (optional)

Evaluation:

- RMSE
- MAE.

👉 Sabse jyada focus Risk formula + seasonal trend pe dena.

## 🧱3️⃣ OOP – Kya Properly Karna Hoga

Yahan tum shine kar sakte ho.

👉 Abstract Base Class for predictors :

Example:

BasePredictor → LinearPredictor inherit kare

**Classes Required:**

### 📌 Data Layer

- DataLoader
- DataPreprocessor

### 📌 Risk Engine

- RiskCalculator
- SeasonalAnalyzer

### 📌 ML Layer

- RiskPredictor
- ModelTrainer

### 📌 Service Layer

- ReportGenerator
- ResourceAllocator

Rule:

- Har class ka single responsibility
- No messy scripts
- Config file separate

OOP pe strong dhyan dena.

## 🌐4️⃣Flask Ka Role

Flask ka kaam ML banana nahi hai.

Flask ka kaam:

- API banana
- Frontend ko data dena
- Admin upload handle karna
- Routes manage karna

Example Routes:

/dashboard
/zone/<id>
/admin/upload
/api/predict

Flask = bridge between ML & Web.

---

## 🗄️5️⃣Database Me Kya Hoga

Simple rakho.

Use:

- SQLite (initial)
- Later PostgreSQL (optional)

Tables:

**zones**

- id
- name
- population
- region

**risk_data**

- zone_id
- date
- traffic
- accidents
- weather
- pollution
- calculated_risk

**predictions**

- zone_id
- month
- predicted_risk

**admin_users**

- id
- username
- password_hash

Focus:

Data schema clean hona chahiye.

## 🎯7️⃣Kis Component Pe Sabse Jyada Dhyan?

Priority order:

1️⃣Risk Formula + Data Logic
2️⃣Seasonal Analysis
3️⃣Clean OOP Structure
4️⃣Dashboard UX
5️⃣Prediction Model

Model sabse last priority hai honestly.

Insight > fancy ML.

## 🔎9️⃣Information Kaha Se Collect Karein?

Very important.

**Data Sources:**

- data.gov.in
- Kaggle (urban traffic datasets)
- WHO pollution data
- IMD weather data

**Research Direction:**

- Google Scholar: “urban risk analysis”
- “seasonal accident trends”
- “municipal risk analytics system”

Tum paper padh ke dekh sakte ho:

- They focus on trend detection
- Seasonal spikes
- Policy recommendation

Ye direction decide karega project ka tone.